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International Mathematics Assessments for Schools

2025/2026 UPPER PRIMARY DIVISION FIRST ROUND PAPER

Time allowed：75 minutes

When your teacher gives the signal, begin working on the problems.

INSTRUCTION AND INFORMATION

GENERAL

1. Do not open the booklet until told to do so by your teacher.
2. No calculators, slide rules, log tables, math stencils, mobile phones or other calculating aids are permitted. Scribbling paper, graph paper, ruler and compasses are permitted, but are not essential.
3. Diagrams are NOT drawn to scale. They are intended only as aids.
4. There are 20 multiple-choice questions, each with 5 choices. Choose the most reasonable answer. The last 5 questions require whole number answers between 000 and 999 inclusive. The questions generally get harder as you work through the paper. There is no penalty for an incorrect response.
5. This is a mathematics assessment, not a test; do not expect to answer all questions.
6. Read the instructions on the answer sheet carefully. Ensure your name, school name and school year are filled in. It is your responsibility that the Answer Sheet is correctly coded.

THE ANSWER SHEET

1. Use only pencils.
2. Record your answers on the reverse side of the Answer Sheet (not on the question paper) by FULLY filling in the circles which correspond to your choices.
3. Your Answer Sheet will be read by a machine. The machine will see all markings even if they are in the wrong places. So please be careful not to doodle or write anything extra on the Answer Sheet. If you want to change an answer or remove any marks, use a plastic eraser and be sure to remove all marks and smudges.

INTEGRITY OF THE COMPETITION

The IMAS reserves the right to re-examine students before deciding whether to grant official status to their scores.

2025 / 2026 UPPER PRIMARY DIVISION FIRST ROUND PAPER

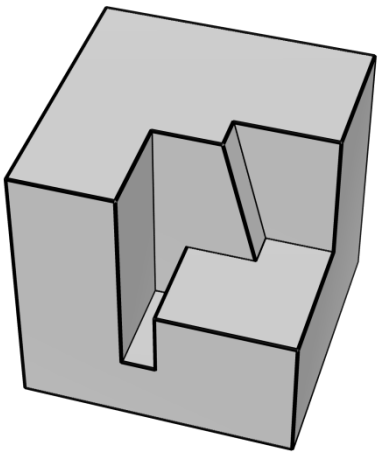
Questions 1-10, 3 marks each

1. Sofia collected some stickers on Monday. On Tuesday, she gave one-third of her stickers plus 6 more to her friend Noah. On Wednesday, she bought 20 additional stickers. On Thursday, she gave half of her stickers to her sister Ava, but Ava returned 4 of them. On Friday, Sofia lost 50% of the stickers she had left. On Saturday, Noah and Ava each returned 10 stickers to Sofia. On Sunday, Sofia divided all her remaining stickers into 10 equal groups, with 14 stickers in each group. How many stickers did Sofia originally collect on Monday?

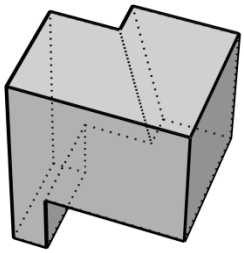
（A）711　　（B）702　　（C）693　　（D）687　　（E）472

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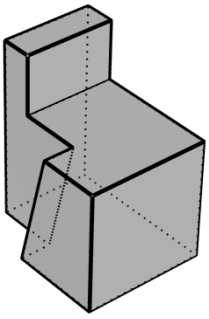
2. Among the options, which solid diagram **CANNOT** interlock with the given diagram shown below to form a solid cube after rotation?



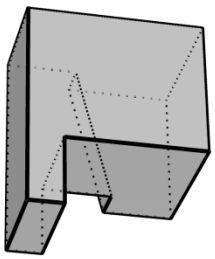
（A）



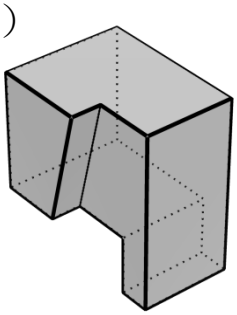
（B）



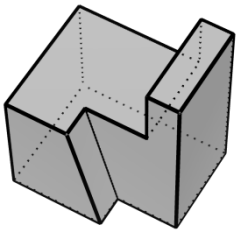
（C）



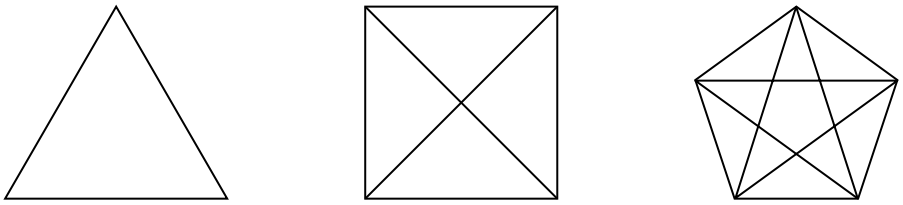
（D）



（E）



3. As shown in the diagram below, an equilateral triangle has no diagonals, a square has 2 distinct diagonals and a regular pentagon has 5 distinct diagonals.

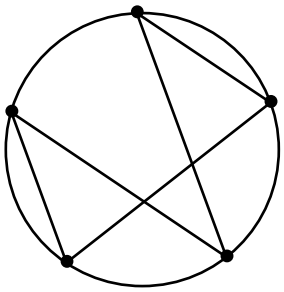


A regular polygon has 54 distinct diagonals. How many sides does it have?

(A) 12 (B) 11 (C) 10 (D) 9 (E) 8

---

4. Five points are evenly spaced along the circumference of a circle. Starting from any point, line segments are drawn to connect each point exactly once before returning to the starting point, forming a 5-sided polygon. One such polygon is shown in the diagram below.

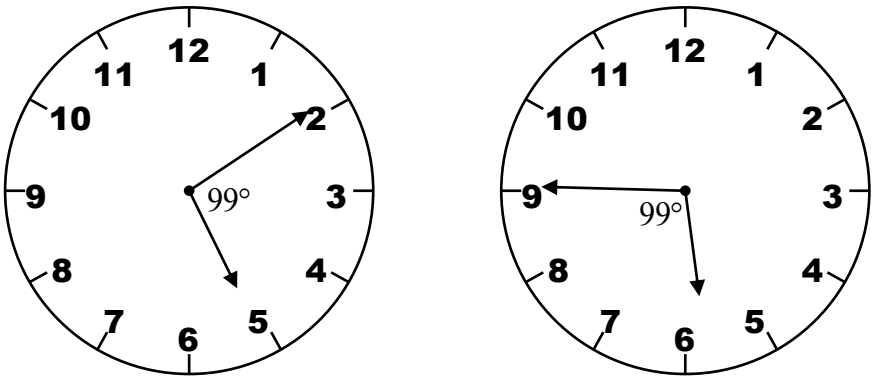


Including the example shown above, how many different polygons can be drawn using these five points?

(A) 3 (B) 4 (C) 10 (D) 11 (E) 12

---

5. When Alex started jogging shortly after 5:00 pm, he noticed that the hour and minute hands of his analogue watch formed a 99° angle. When he finished jogging before 6:00 pm, he noticed that the hands again formed a 99° angle. How many minutes did Alex jog?

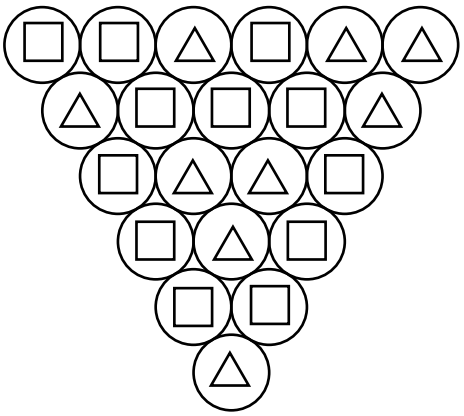


Started jogging Finished jogging

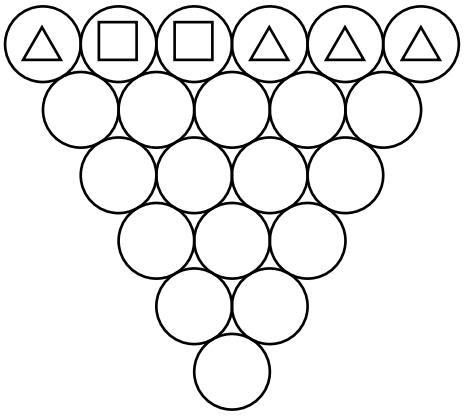
(A) 36 (B) $36\frac{2}{11}$ (C) $36\frac{4}{11}$ (D) $36\frac{6}{11}$ (E) 37

---

6. Each circle in the diagram below contains either a □ or a △, arranged according to a specific rule.



Six symbols (□ or △) have already been placed in the circles on the top row of the diagram below. Following the same rule as above, the remaining circles will be filled with either a □ or a △. Once the diagram is filled in correctly, how many △'s will be placed in total, including those in the top row?



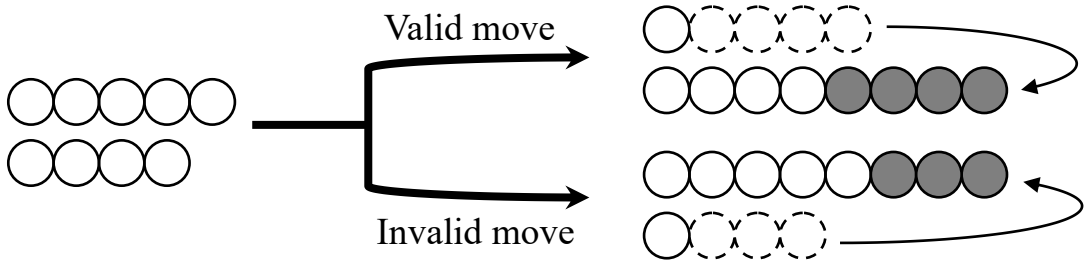
（A）10　　（B）11　　（C）12　　（D）13　　（E）14

---

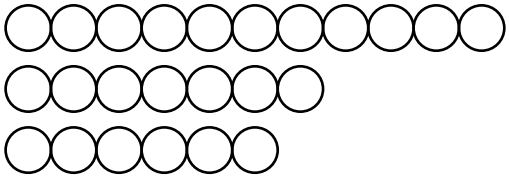
7. At an international conference, 120 headsets were prepared, each requiring 4 batteries to power them. It was estimated that only 75% of the headsets would be in use at any given time. However, due to a mistake, the technicians prepared a newer headset model that requires 6 batteries each, while still assuming that only 75% of them would be used simultaneously. How many additional batteries are needed to meet the estimated demand compared to before?

（A）240　　（B）180　　（C）60　　（D）30　　（E）24

8. Some marbles are arranged in several rows. In each turn, a number of marbles is moved from one row to another such that the number of marbles moved equals the number of marbles in the target row just before the move. For example, there are 9 marbles that are arranged in 2 rows, as shown in the diagram below. One valid move and one invalid move are also shown.



Now, as shown in the diagram below, 24 marbles are arranged in three rows: 11 in the first row, 7 in the second row and 6 in the third row. Using the same procedure as above, what is the minimum number of turns required such that each row contains exactly 8 marbles?



(A) 2 (B) 3 (C) 4
(D) 5 (E) Greater than or equal to 6

---

9. Alice and Billy live in the same apartment building. Units 0 to 9 are on the 1st floor, units 10 to 19 are on the 2nd floor, units 20 to 29 are on the 3rd floor, and so on. Alice's unit number is on the floor that matches Billy's unit number. If the sum of their unit numbers is 255, what is the unit number of Alice's apartment?

(A) 230 (B) 231 (C) 232 (D) 251 (E) 252

---

10. How many ordered pairs of positive integers (m, n) have a least common multiple (LCM) equal to 2025?

(A) 15 (B) 29 (C) 37 (D) 38 (E) 45

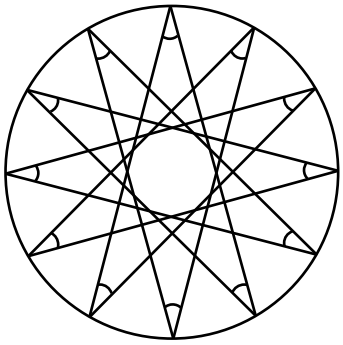
Questions 11-20, 4 marks each

11. A store sells T-shirts, shirts and jackets. The price of a jacket is equal to the sum of the prices of a T-shirt and a shirt. If the price of a shirt were increased by 80%, its new price would be equal to the sum of the prices of a T-shirt and a jacket. By what percentage would the price of a T-shirt need to increase to equal the sum of the prices of a shirt and a jacket?

(A) 50% (B) 60% (C) 160% (D) 500% (E) 600%

---

12. Twelve points are evenly spaced on the circumference of a circle. These points are connected in a specific pattern to form a star polygon, as shown in the diagram below. What is the sum of the 12 interior angles, in degrees, of the star polygon?



（A）90°　（B）120°　（C）180°　（D）360°　（E）720°

---

13. Alice, Bob and Carol are working together to complete a task. Assume each person works at a constant rate and is each responsible for one-third of the total work. Alice starts at 9:00 am, Bob starts at 10:00 am and Carol starts at 10:30 am. They all take a break at 12:00 pm, by which they have already completed half of the total work. After lunch, they resume working at 1:00 pm. If Alice finishes her portion at 3:00 pm and Bob finishes his portion at 4:00 pm, at what time will Carol complete her portion of the task?

（A）1:50 pm　（B）2:00 pm　（C）2:30 pm　（D）3:30 pm　（E）4:30 pm

---

14. Let n be the smallest positive integer such that $n+3n+5n+7n+9n+11n$ is a number whose digits are all the same. What is the sum of the digits of n?

（A）27　（B）36　（C）37　（D）38　（E）72

---

15. Three fair dice, whose colours are red, yellow and blue, each with faces numbered from 1 through 6, are all thrown at the same time. How many different outcomes are there where the 3 dice rolls sum to a perfect square?

Note: Dice with the same numbers but have different colours are considered distinct outcomes. For example, (1, 2, 3) and (2, 3, 1) are different outcomes, where the ordered triple (r, y, b) represents the numbers appearing on top of the red, yellow and blue dice, respectively.

（A）9　（B）28　（C）31　（D）34　（E）39

---

16. There are four positive integers a, b, c and d. If the greatest common divisors (GCD) of all six pair-wise distinct pairs are 1, 2, 3, 4, 6 and 15, what is the least possible value of $a\times b\times c\times d$?

（A）43200　（B）24000　（C）21600　（D）10800　（E）7200

---

17. Alex, Ben, and Carl each have a certain number of marbles, with one of them having exactly 25 marbles. They exchange marbles in the following manner:

- First Exchange: Alex receives marbles from Ben equal to Alex's original number.
- Second Exchange: Ben receives marbles from Carl equal to Ben's number just before this exchange.
- Third Exchange: Carl receives marbles from Alex equal to Carl's number before the exchange.

After these three exchanges, they each have the same number of marbles. How many marbles did Carl have initially?

(A) 20 (B) 25 (C) 30 (D) 40 (E) 45

---

18. The first two terms of a sequence are 2 and 7. Starting from the third term onward, each term is the sum of all terms that come before it. Which of the following numbers is **NOT** a term in the sequence?

(A) 9216 (B) 6912 (C) 2304 (D) 576 (E) 72

---

19. Andy has 12 cards with the following statements written on them:

"There is exactly one false statement to the left of this card."

"There are exactly two false statements to the left of this card."

…

"There are twelve false statements to the left of this card."

Andy arranges all cards in a row, from left to right, in some order.

What is the largest possible number of statements that can be true?

(A) 1 (B) 2 (C) 4 (D) 6 (E) 10

---

20. The number 11^{2025} is written on the board in its decimal representation. The sum of its digits is calculated repeatedly, by summing the digits of each result, until a single digit number is obtained. What is the value of the final digit?

(A) 0 (B) 2 (C) 4 (D) 6 (E) 8

Questions 21-25, 6 marks each

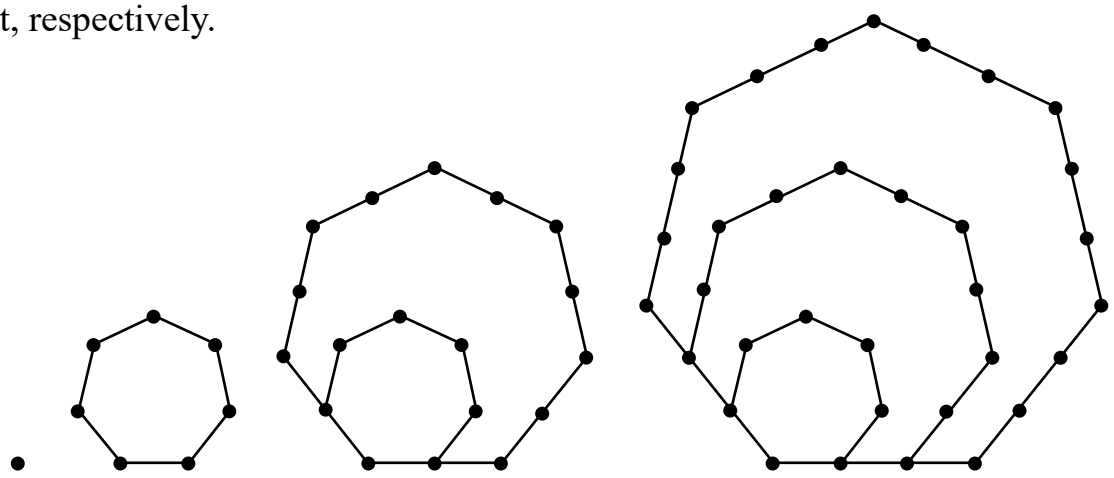
21. Victor is struggling to recall his girlfriend's phone number. He knows it is a 10-digit number. He only recalls the following:

- It starts with 07.
- The third digit is either 4 or 5.
- The next six digits consist of three 1s, two 2s and one 3, but he doesn't remember the order.
- The last digit is a prime number.

At most how many tries does Victor have to make to guess the correct number?

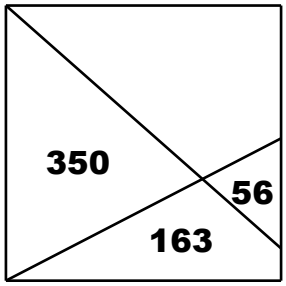
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22. As shown in the diagrams below, the first four 7-gonal numbers are 1, 7, 18 and 34, which correspond to the number of black dots in each diagram from left to right, respectively.

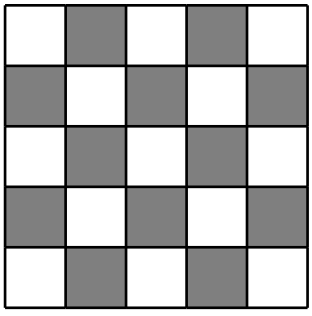


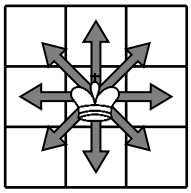
If 616 is the mth 7-gonal number, then what is the value of m?

23. A square garden is divided into four regions by two straight lines that pass through two adjacent corners and meet at a point inside the square, as shown in the diagram below. If the areas of three of the regions are labelled as 350 m^2, 163 m^2 and 56 m^2, what is the area, in m^2, of the fourth region?



24. A king chess piece moves on a 5×5 chessboard with alternating black and white squares, such that it passes through each square exactly once. What is the minimum number of colour changes the king must make along its path?
(Note: The king chess piece can move as shown in the diagram on the right.)





25. Among the positive integers less than 1000, how many are multiples of 7 and have a decimal representation that contains at least one digit 5?

* * *